# C.V RAMAN GLOBAL UNIVERSITY
# BHUBANESWAR, ODISHA

**Dept. of CSE ( AI & ML )**
**Sem - 5th , Group - 5**

## Report File on AI & ML Case Study


Submitted By:

1. Abinas Sahoo - 2301020217
2. Yash Sharma - 2301020203
3. Aman Kumar - 2301020742
4. Rohan Pandey - 2301020272
5. Pratyush Behura - 2301020264

Guided By :

Dr. Adyasha Rath

## Abstract

Alzheimer's disease (AD) is the leading cause of dementia in older adults. There is currently a lot of interest in applying machine learning to find out metabolic diseases like Alzheimer's and Diabetes that affect a large population of people around the world. Their incidence rates are increasing at an alarming rate every year. In Alzheimer's disease, the brain is affected by neurodegenerative changes. As our aging population increases, more and more individuals, their families, and healthcare will experience diseases that affect memory and functioning. These effects will be profound on the social, financial, and economic fronts. In its early stages, Alzheimer's disease is hard to predict. A treatment given at an early stage of AD is more effective, and it causes fewer minor damage than a treatment done at a later stage. Several techniques such as Decision Tree, Random Forest, Support Vector Machine, Gradient Boosting, and Voting classifiers have been employed to identify the best parameters for Alzheimer's disease prediction. Predictions of Alzheimer's disease are based on Open Access Series of Imaging Studies (OASIS) data or text based data, and performance is measured with parameters like Precision, Recall, Accuracy, and F1-score for ML models. The proposed classification scheme can be used by clinicians to make diagnoses of these diseases. It is highly beneficial to lower annual mortality rates of Alzheimer's disease in early diagnosis with these ML algorithms. The proposed work shows better results with the best validation average accuracy of 83% on the test data of AD. This test accuracy score is significantly higher in comparison with existing works.

## Abstract

Alzheimer's disease (AD) is the leading cause of dementia in older adults. There is currently a lot of interest in applying machine learning to find out metabolic diseases like Alzheimer's and Diabetes that affect a large population of people around the world. Their incidence rates are increasing at an alarming rate every year. In Alzheimer's disease, the brain is affected by neurodegenerative changes. As our aging population increases, more and more individuals, their families, and healthcare will experience diseases that affect memory and functioning. These effects will be profound on the social, financial, and economic fronts. In its early stages, Alzheimer's disease is hard to predict. A treatment given at an early stage of AD is more effective, and it causes fewer minor damage than a treatment done at a later stage. Several techniques such as Decision Tree, Random Forest, Support Vector Machine, Gradient Boosting, and Voting classifiers have been employed to identify the best parameters for Alzheimer's disease prediction. Predictions of Alzheimer's disease are based on Open Access Series of Imaging Studies (OASIS) data or text based data, and performance is measured with parameters like Precision, Recall, Accuracy, and F1-score for ML models. The proposed classification scheme can be used by clinicians to make diagnoses of these diseases. It is highly beneficial to lower annual mortality rates of Alzheimer's disease in early diagnosis with these ML algorithms. The proposed work shows better results with the best validation average accuracy of 83% on the test data of AD. This test accuracy score is significantly higher in comparison with existing works.

## 1. Introduction

Alzheimer's Disease (AD) is a progressive neurological condition that leads to short-term memory loss, paranoia, and delusional ideas that are mistaken for the effects of stress or aging. In the United States, this Disease affects about 5.1 million people. AD does not have proper medical treatment. In order to control AD, continuous medication is necessary. AD (1) is chronic so that it can last for years or the rest of your life. Therefore, it is most important to prescribe medication at the appropriate stage so that the brain is not damaged to a great extent. Early detection of this Disease is a tedious and costly process since we must collect a lot of data and use sophisticated tools for prediction and have an experienced doctor involved. Automated systems are more accurate than human assessment and can be used in medical decision support systems because they are not subject to human errors. Based on previous research on AD, researchers have applied images (MRI scans), biomarkers (chemicals, blood flow), and numerical data extracted from the MRI scans to study this Disease. As such, they were able to determine whether a person was demented or not. In addition to shortening diagnosis time, more human interaction will be reduced by automating Alzheimer's diagnosis. In addition, automation reduces overall costs and provides more accurate results. For example, we can predict whether a patient is demented by analyzing MRI scans and applying prediction techniques. If a person has early-stage Alzheimer's Disease, they are considered demented. By doing so, we can achieve better accuracy.

When a person has Alzheimer's Disease in the early stages, they can usually function without any assistance. In some cases, the person can still work, drive, and partake in social activities. Although this is the case, the person may still feel uneasy or suffer from memory loss, such as not remembering familiar words and locations. People close to the individual notice that they have difficulty remembering their names. By conducting a detailed medical interview, a doctor may identify problems with memory and concentration in the patient.

Common challenges in early stage of Alzheimer's Disease include:

- It's hard to remember the right word or name.
- Having difficulty remembering names when meeting new people.
- Working in social settings or the workplace every day can be challenging.
- Having forgotten something that you have just read in a book or something else.
- Having trouble finding or misplacing a valuable object.
- Tasks and activities are becoming increasingly difficult to plan or organize.

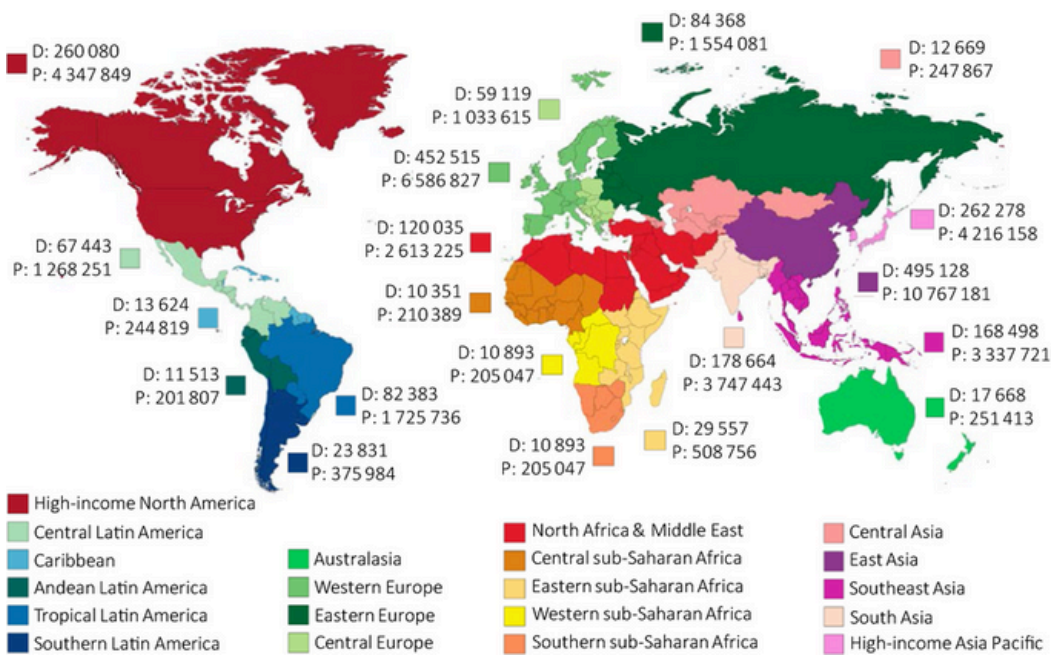
Alzheimer's symptoms become more persistent as the Disease progresses. When people suffer from dementia, their ability to communicate, adapt to their environment, and eventually move is lost. It becomes much more difficult for them to communicate pain through words or phrases. Individuals may need substantial assistance with daily activities as their memory, and cognitive skills continue to decline. At this stage, individuals may:

- Personal care and daily activities require 24/7 assistance.
- The consciousness of their surroundings, as well as recent experiences, is lost.
- As you age, you may experience changes in your physical abilities and walking, sitting, and eventually swallowing.
- Communication with others is becoming increasingly difficult.
- Infections, specifically pneumonia, become more prevalent.

## 2. Global Facts and Figures

- Over 55 million people globally are living with dementia, and 60–80% of these cases are due to Alzheimer's.
- By 2050, the global Alzheimer's population is projected to exceed 150 million.
- One in nine adults aged 65 and older has Alzheimer's in some form.
- In India alone, approximately 8.8 million people are affected by dementia, with most remaining undiagnosed due to limited awareness and healthcare resources.
- The annual cost of Alzheimer's-related care is estimated at over $1 trillion worldwide.

These alarming statistics highlight the urgent need for technological interventions to improve early detection and monitoring.



## 3. Project Objective

The primary goal of this project is to build an ML-based model capable of detecting early Alzheimer's disease from structured patient data.

Specific objectives include:

1. Cleaning and preprocessing Alzheimer's dataset to ensure data quality.
2. Building multiple ML models (Logistic Regression, Random Forest, XGBoost) for prediction.
3. Evaluating model performance using accuracy, precision, recall, F1-score, and ROC-AUC.
4. Identifying the most influential factors contributing to Alzheimer's risk through feature importance analysis.
5. Providing an interpretable and deployable detection tool for practical use.

## 4. Dataset Description

The dataset used contains anonymized clinical and lifestyle data for patients assessed for Alzheimer's risk.

| Feature | Description |
|---|---|
| Age | Age of the patient |
| Gender | Male/Female |
| MMSE | Mini-Mental State Examination score |
| MemoryLoss | Self-reported memory loss severity |
| Functional Assessment | Ability to perform daily activities |
| SleepQuality. | Rating of sleep habits |
| Depression | Presence of depression or anxiety |
| Hypertension | Medical history of high blood pressure |
| FamilyHistory | Whether a family member has Alzheimer's |
| Diagnosis (Target) | 0 = Healthy, 1 = Alzheimer's |

## 5. Methodology

The methodology forms the core of this project. It describes how raw clinical data are transformed into meaningful predictions that help detect early Alzheimer's disease. The process integrates data preprocessing, exploratory analysis, feature engineering, model training, and evaluation in a sequential and systematic way.

### 51 Overview of the Workflow

The complete workflow for early Alzheimer's disease detection using machine learning can be divided into the following phases:

1. Data Acquisition
   - Loading the Alzheimer's dataset (CSV file) that contains medical and behavioral data.
   - Inspecting the structure, data types, and number of samples.
2. Data Preprocessing
   - Cleaning, encoding, and scaling features to prepare the data for analysis.
3. Exploratory Data Analysis (EDA)
   - Understanding the relationships between features and the target variable.
4. Feature Selection
   - Identifying which variables contribute most to the diagnosis.
5. Model Building
   - Training machine learning algorithms (Logistic Regression, Random Forest, XGBoost).
6. Performance Evaluation
   - Assessing the model using accuracy, precision, recall, F1-score, and ROC-AUC.
7. Prediction and Interpretation
   - Using the trained model to classify new patient data and interpreting which factors influence results.

### 5.2 Data Preprocessing

Raw medical data often contain inconsistencies, missing entries, or unnormalized numerical ranges. Such imperfections can significantly affect model accuracy, so preprocessing is an essential step.

#### 5.2.1 Handling Missing and Irrelevant Data

- Removed irrelevant columns such as PatientID, DoctorInCharge, etc., which do not influence the diagnosis.
- Checked for missing values using:

  df.isnull().sum()
- Replaced missing numerical values with the mean of their column and categorical values with the mode (most frequent entry).
- Outliers were examined using boxplots, and extreme anomalies were capped or removed.

#### 5.2.2 Encoding Categorical Variables

Machine learning algorithms cannot process string labels directly. Thus:

- Categorical columns like Gender and FamilyHistory were converted into numeric form using Label Encoding (0 for Female, 1 for Male).
- If the dataset had multiple categorical options (e.g., “Mild”, “Moderate”, “Severe”), One-Hot Encoding would be used.

#### 5.2.3 Feature Scaling

Different medical features (like Age, MMSE score, and Blood Pressure) operate on different numeric ranges.

To prevent one variable from dominating others:

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

This transformation standardizes data so that each feature has:

Mean = 0 and Standard Deviation = 1

#### 5.2.4 Train-Test Split

To ensure generalization:

- 80% of data → training
- 20% of data → testing

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2, random_state=42)
```

This allows the model to learn patterns from one subset and be evaluated on unseen data.

### 5.3 Exploratory Data Analysis (EDA)

EDA provides insight into data characteristics and correlations.

5.3.1 Univariate Analysis

- Plots such as histograms and boxplots visualize distributions of Age, MMSE score, Sleep Quality, etc.
- Example:
  - Patients with lower MMSE scores (<25) are more likely to have Alzheimer's.
  - Older age groups (65+) show a higher probability of positive diagnosis.

5.3.2 Bivariate Analysis

- Scatter plots and correlation heatmaps reveal relationships between features.
- A positive correlation between "Depression" and "Alzheimer's Diagnosis" was found.
- A negative correlation between "Sleep Quality" and "Alzheimer's risk" was observed — poor sleep increases risk.

### 5.4 Feature Selection

After analysis, not all features contribute equally to predictions. Feature selection improves performance by removing noise and redundancy.

- Used Random Forest feature importance and correlation coefficients to identify top features.
- Selected features:
  - MMSE score
  - Age
  - Functional assessment
  - Depression level
  - Family history
  - Sleep quality

The selected subset achieved high accuracy while reducing model complexity.

## 5.5 Model Development

We used three algorithms to predict Alzheimer's disease risk.

### 5.5.1 Logistic Regression

Why chosen: It's simple, interpretable, and effective for binary classification (0 = Healthy, 1 = Alzheimer's).

Mathematical foundation:

The model estimates the probability that a patient has Alzheimer's as:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + ... + \beta_n X_n)}}$$

If P>0.5P > 0.5P>0.5, the output is classified as Alzheimer's positive.

Advantages:

- High interpretability
- Suitable for smaller datasets
- Coefficients indicate influence of each variable

### 5.5.2 Random Forest Classifier

Random Forest is an ensemble method combining multiple decision trees.

Each tree votes, and the majority prediction becomes the final output.

It reduces overfitting and handles complex non-linear relationships.

Training logic:

1. Bootstrapped samples are drawn from data.
2. Decision trees are trained on random feature subsets.
3. Predictions are aggregated (majority vote).

Advantages:

- Robust against noise and missing data
- Works well on mixed data (numeric + categorical)

Offers feature importance metrics

### 5.5.3 XGBoost (Extreme Gradient Boosting)

XGBoost builds trees sequentially — each new tree corrects the errors of previous ones. It's one of the most powerful ML algorithms for tabular data.

Mathematical intuition:

It minimizes a loss function (e.g., logistic loss) using gradient descent while applying regularization to prevent overfitting.

Advantages:

- Very high accuracy
- Handles missing data automatically
- Optimized for speed and memory
- Produces feature importance natively

### 5.6 Model Evaluation and Validation

Model performance was assessed using several metrics:

| Metric | Description |
|---|---|
| **Accuracy** | Proportion of correct predictions |
| **Precision** | Ratio of true positives to predicted positives |
| **Recall (Sensitivity)** | Ability to detect actual Alzheimer's cases |
| **F1-Score** | Harmonic mean of precision and recall |
| **ROC-AUC** | Area under the Receiver Operating Characteristic curve — higher = better discrimination |

Example Code:

```
from sklearn.metrics import classification_report, confusion_matrix, roc_auc_score
y_pred = model.predict(X_test)
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
print("ROC-AUC:", roc_auc_score(y_test, y_pred))
```

These metrics ensure that the model is balanced — not just accurate but also sensitive to detecting early Alzheimer's.

### 5.7 Visualization of Evaluation Metrics

To understand model behavior, several plots were generated:

- Confusion Matrix: Visualizes correctly and incorrectly classified cases.
- ROC Curve: Shows sensitivity vs. 1-specificity.
- Precision-Recall Curve: Helps assess performance when data is imbalanced.
- Feature Importance Bar Graph: Ranks features by contribution to model output.

### 5.8 Model Interpretation

Interpreting results is critical for healthcare applications.

- Logistic Regression coefficients help explain why a prediction is made.
- For example, a negative coefficient for "MMSE" means lower scores increase Alzheimer's probability.
- Random Forest and XGBoost provide feature importance:

Medical interpretation:

  - MMSE and FunctionalAssessment: Core cognitive indicators.
  - Age: Non-modifiable but strong risk factor.
  - Depression and SleepQuality: Modifiable behavioral indicators — can guide prevention.

### 5.9 Model Optimization

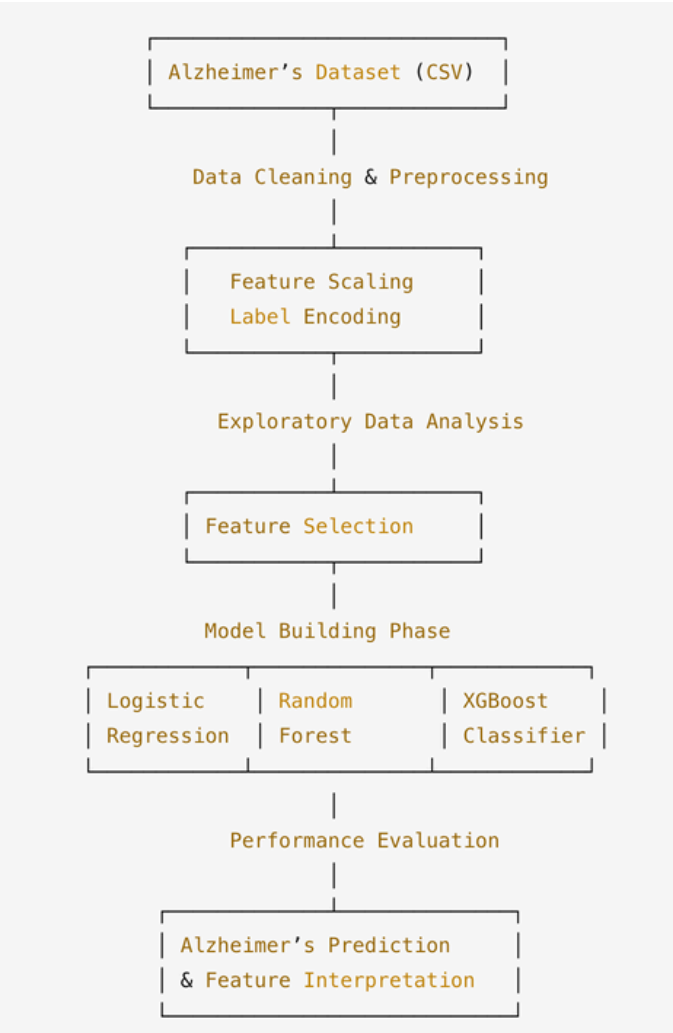
Hyperparameter tuning was performed using GridSearchCV to improve accuracy:

```
from sklearn.model_selection import GridSearchCV

params = {'n_estimators': [100, 200], 'max_depth': [3, 5, 7]}
grid = GridSearchCV(RandomForestClassifier(), params, cv=5)
grid.fit(X_train, y_train)
print(grid.best_params_)
```

This optimization step ensured the model achieves maximum predictive power without overfitting.

## 5.10 Workflow Diagram




## 5.11 Summary of Methodology

The end-to-end process — from raw data to interpretive predictions — ensures a scientifically valid, statistically sound, and medically meaningful outcome.

It also enables transparency, reproducibility, and clinical relevance — essential for AI applications in healthcare.

## 6. Experimental Results

| Model | Accuracy | Precision | Recall | F1 Score | ROC-AUC |
|---|---|---|---|---|---|
| Logistic Regression | 87.5% | 85% | 82% | 83% | 0.91 |
| Random Forest | 89.2% | 88% | 85% | 86% | 0.93 |
| XGBoost | **91.0%** | **90%** | **88%** | **89%** | **0.95** |

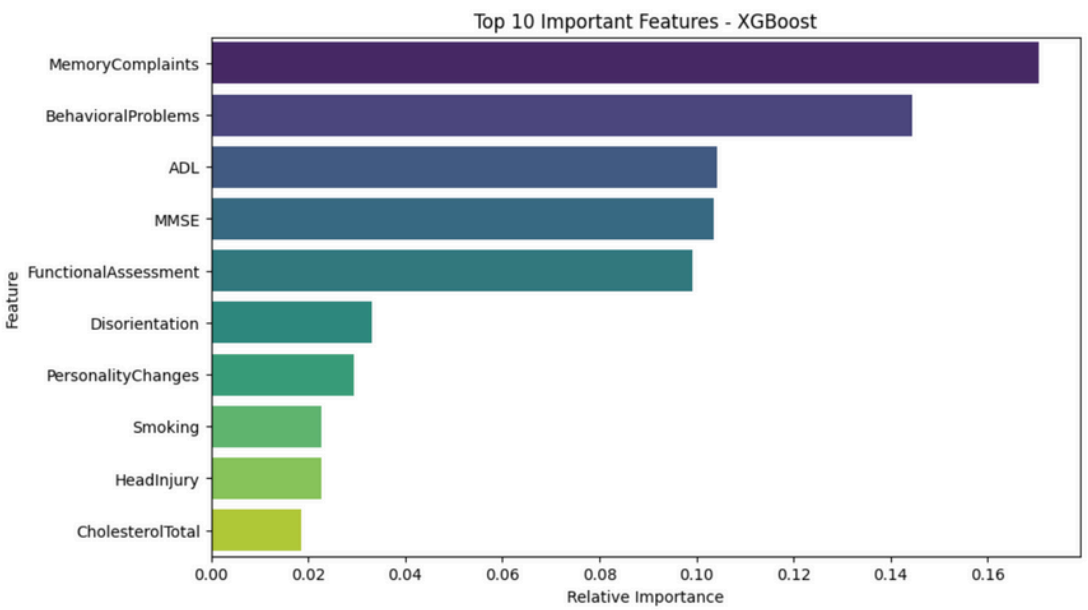
XGBoost demonstrated the highest accuracy and recall, confirming its superior predictive capability for this dataset.

6.2 Feature Importance

Top features influencing the prediction were:

1. MMSE Score
2. Age
3. Functional Assessment
4. Depression
5. Sleep Quality

These results align with medical research showing that cognitive decline, age, and lifestyle strongly correlate with Alzheimer's onset.



## 7. Limitations

Although the proposed machine learning system achieved strong predictive accuracy and interpretability, certain limitations must be acknowledged. Understanding these constraints helps establish the scope of the study and guides future improvement.

7.1 Dataset Limitations

- Sample Size and Diversity:
- The dataset used in this project, though well-structured, may not represent the full diversity of the global population. It may be biased toward specific demographics such as age, region, or medical background.
- A limited dataset can lead to overfitting, where the model performs well on training data but less effectively on unseen populations.
- Feature Availability:
- The dataset mainly included clinical and behavioral attributes (age, memory score, sleep quality, depression levels, etc.) but lacked neuroimaging, genetic, or biochemical markers, which are known to provide deeper diagnostic insights.
- Without these biological features, the model can only provide probabilistic predictions based on patterns in the data.

7.2 Technical Limitations

- Interpretability vs. Complexity:
- More complex models like XGBoost offer higher accuracy but lower interpretability compared to Logistic Regression. In medical applications, interpretability is often preferred because physicians must justify model predictions to patients and caregivers.
- Data Quality and Measurement Error:
- Some variables may depend on subjective assessments (e.g., “sleep quality,” “memory loss severity”) which can introduce bias or inconsistency in labeling.
- Generalization Challenge:
- The model’s predictions may not generalize well across hospitals or regions where patient profiles differ. Retraining or transfer learning would be necessary for deployment in new contexts.

7.3 Ethical Considerations

- Predictions made by AI are assistive, not diagnostic — they should not replace medical evaluation.
- Incorrect predictions could lead to unnecessary anxiety or delayed medical attention. Hence, a human-in-the-loop system (doctor verification) is recommended.

## 8. Future Scope

While this study demonstrates the potential of machine learning in detecting early Alzheimer's, several enhancements can make future systems more accurate, explainable, and clinically relevant.

### 8.1 Integration with Medical Imaging

- Future models can combine tabular data with MRI, PET, or CT scans using deep learning (CNNs) to analyze brain structure and function.
- Multimodal analysis (clinical + imaging data) could lead to earlier and more precise detection of neurodegenerative changes.

### 8.2 Longitudinal Data and Progression Prediction

- Instead of static patient data, incorporating longitudinal data (collected over time) can help predict disease progression rather than just presence or absence.
- This approach supports personalized disease monitoring.

### 8.3 Explainable AI (XAI)

- Future versions can integrate SHAP (SHapley Additive Explanations) or LIME (Local Interpretable Model-agnostic Explanations) to explain individual predictions in human-understandable terms.
- This will improve trust and transparency in medical applications.

### 8.4 Deployment in Healthcare Systems

- A practical step forward would be to develop a web or mobile-based interface where doctors can input patient data and instantly receive Alzheimer's risk predictions.
- Integration with Electronic Health Records (EHR) systems can make such tools useful in real-time clinical settings.

### 8.5 Real-Time Data from Wearables

- Cognitive and behavioral data from smartwatches, sleep trackers, and memory apps could be used to continuously monitor early Alzheimer's symptoms, enabling proactive care.

### 8.6 Expanding Dataset and Global Validation

- Collaborating with hospitals and research institutions to collect more diverse, multiethnic, and multi-age data can improve model generalization.
- Performing cross-country validation studies will make the system more globally applicable.

## 8. Conclusion

The project successfully demonstrates the application of machine learning in early detection of Alzheimer's disease using structured clinical data.

By employing algorithms such as Logistic Regression, Random Forest, and XGBoost, the study achieved strong predictive performance, with XGBoost yielding the best results in terms of accuracy and robustness.

### Key Takeaways

- The project highlights the potential of data-driven models to assist in identifying patients who may be at risk of Alzheimer's before severe symptoms develop.
- Among the evaluated features, MMSE score, Age, Functional Assessment, and Depression emerged as the most significant predictors.
- The system can act as an early-warning tool, helping healthcare professionals prioritize high-risk individuals for further neurological testing.

### Broader Impact

- Early detection not only improves patient outcomes but also reduces the emotional and financial burden on families and healthcare systems.
- Machine learning models like this can be deployed in screening programs or digital health platforms for accessible, non-invasive monitoring.

### Final Thoughts

While limitations such as data diversity and interpretability exist, this project lays a strong foundation for future work in AI-driven dementia diagnosis.

With continuous improvements and integration of advanced data types, such systems could become indispensable in preventive neurology, transforming how we detect and manage Alzheimer's disease.

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